

homework-2

Andrew Barefoot

Question 1

```
install.packages("aRtsy")
```

Install.packages is how you download a package from a repo and you only have to do it once. When you library something it loads it into the current session. That is something you have to do every R session.

Packages

Question 2

```
library(aRtsy)
```

The Collatz conjecture is a famous mathematics problem that is unsolved. It states that no matter what positive integer you start with, this sequence will always eventually reach 1.

$$f(n) = \begin{cases} n/2 & \text{if } n \equiv 0 \pmod{2} \\ 3n + 1 & \text{if } n \equiv 1 \pmod{2} \end{cases}$$

Question 3

part a

This function generates artwork based on the sequence. It takes the number in the function and makes that number of random lines.

part b

You get a different output every time because it uses R's built in number generator instead of a fixed drawing for each number.

This is troublesome in the .qmd file because the picture may look different every time you render it. This makes it non-reproducible for anyone else.

Set.seed is an integer vector, containing the random number generator state for random number generation in R. If you use a seed the randomness will produce the exact same random result every time it is run.

```
set.seed(8)

canvas_collatz(
  colors = c("blue"),
  n = 10
)
```



Question 4

```
canvas_collatz(  
  colors = c("blue"),  
  n = 1:20  
)
```



This code is using a random number of lines to make its artwork. This is because last question the function had $n = 10$ which tells R to make the artwork with exactly 10 lines, where as this function uses a vector of $n = 1:20$, which makes the code choose a random number of lines within the vector to make its artwork.

I would say yes because there is still a lot of randomness with the vector that `set.seed(1)` would help to control.